Roll No. ..............................

# TCS–403

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

MICROPROCESSORS

Time : Three Hours

Maximum Marks : 100

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) List the registers present in 8086 and draw the format of flag register of 8086. (CO1, CO3)

   (b) Bring out the architectural and signal differences between 8086 and 8088. Explain status signals of 8086. (CO1, CO3)

   (c) Write a program in 8086 to find factorial of a number stored at offset 2050 and store the result at starting offset 3050. (CO1, CO3)

2. (a) Draw the instruction cycle of instruction JMP 2050. (CO1, CO2)

   (b) Explain concept of memory interfacing with example to interface 2KB RAM with microprocessor 8085. (CO1, CO2)

*P. T. O.*

(c) Write a program in 8085 which determines smallest number in an array of 50 numbers. (CO1, CO2)

3. (a) Explain different types of interrupts of 8086. (CO2, CO3)

(b) Explain relative addressing mode and based + indexed + displacement addressing mode in 8086 with two examples. Also defermine physical memory address in each example. (CO2, CO3)

(c) Explain the meaning of the following 8086 instructions : (CO3, CO5)

(i) XLAT

(ii) STOSB

(iii) AAM

(iv) LAHF

4. (a) Write a program in 8086 to convert a 16-bit binary number into gray number. Also determine addressing mode of each instruction used in this program. (CO3, CO5)

(b) Write a program in 8086 which determine sum of digits of a 16-bit number. (CO3,CO5)

(c) Draw and explain the block diagram of 8254. (CO3, CO5)

5. (a) What is the application of 8251 USART ? Explain its block diagram.

(b) Draw the control word format of 8255 in input-output mode. Write a program in 8085 which reverse contents of port B and store the result at port A and port C. Address of control register is 83H, PA 80H, PB = 81H and PC 82H. (CO6)

(c) Write a program in 8085 which interface ADC 0800 with 8085. (CO6)

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# TCS–402

## B. TECH. (CSE) (FOURTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023
### FINITE AUTOMATA AND FORMAL LANGUAGES

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 20 marks.

1. (a) Given the CFG below convert it into Chomsky Normal Form (CNF) : (CO3)

$$S \rightarrow aXbX$$
$$X \rightarrow aY \mid bY \mid \varepsilon$$
$$Y \rightarrow X \mid c$$

(b) What is Turing machine ? Discuss about the various types of Turing machine (at least *five*) along with language of it. (CO5)

(c) Implement a Turing machine for language L given below : (CO5)

$$L = \{0^n 1^n 2^n \mid n >= 1\}, \Sigma = (0, 1, 2)$$

2. (a) Implement a push down automata for language L given below : (CO4)

$L = \{WW^r \mid W \in (0, 1)^*$ and r represents the reverse$\}$ for e.g.

If W = 011, then $WW^r$ will be 011110. (CO4)

*P. T. O.*

(b) Using the pumping for CFL, prove that the given language L is not CFL : (CO3)

$$L = \{WW \mid W \in (0, 1)^*\}$$

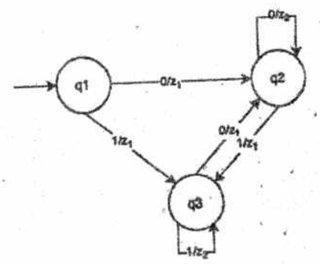
for e.g. If W = 011, then WW will be 011011.

(c) Draw the minimum state DFA for the given language L below : (CO1)

L = {all strings having length at least 2 and even number of 0's}, $\Sigma = (0, 1)$ for e.g. 0011, 010, 100 all are accepted strings but $\varepsilon$ is not acceptable.

3\. (a) Design NFA for regular expression (10)*1+10(=+1)*1+(010+110)*. (CO1)

(b) Construct Moore machine from the given Mealy machine. (CO2)




(c) Construct a Moore machine that takes radix 4 number as input and products residue modulo '5' as output : (CO2)

$$\Sigma = (0, 1, 2, 3)\ \Delta = \{0, 1, 2, 3, 4\}$$

4\. (a) Construct a regular expression to generate all strings having NO two a's and NO two b's are together. $\Sigma = (a, b)$ for e.g. $\varepsilon$, a, b, ab, aba, bab ............... are acceptable but aab or bba are NOT acceptable. (CO2)

(b) Using the pumping lemma for Regular Language, prove that the given language L is not regular : (CO1)

L $L = \{W \times W^r \mid W \in (0, 1)^*$, r represents the reverse and X is a special symbol which represents middle of a string} for e.g.

If W = 011 then $W \times F^r$ will be $011 \times 110$.

(c) Construct a push down automata for the language given below :

$$L = \{a^p b^q c^r d^s \mid, \text{ where either } p = = r \text{ or } q = = s\}.$$

It is guaranteed that PDA exists for the given language. (CO4)

5. (a) Discuss about the following closure properties of the CFL with suitable example to support your statement : (CO6)

(i) Result of union of two CFLs

(ii) Result of intersection of two CFSs

(iii) Complement of à CFL.

(b) Construct a Transducer Turing machine which act as a copier of the given string. For e.g. (CO5)

| | | | | | |
|---|---|---|---|---|---|
| B | 1 | 0 | 1 | 1 | B |

Initial Turing Tape

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| B | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | B |

Turing Tape after copy

(c) What is Universal Turing Machine (UTM) ? Explain with a suitable example. (CO6)

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# TCS–403

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

MICROPROCESSORS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) List the registers present in 8086 and draw the format of flag register of 8086. (CO1, CO3)

   (b) Bring out the architectural and signal differences between 8086 and 8088. Explain status signals of 8086. (CO1, CO3)

   (c) Write a program in 8086 to find factorial of a number stored at offset 2050 and store the result at starting offset 3050. (CO1, CO3)

2. (a) Draw the instruction cycle of instruction JMP 2050. (CO1, CO2)

   (b) Explain concept of memory interfacing with example to interface 2KB RAM with microprocessor 8085. (CO1, CO2)

*P. T. O.*

(c) Write a program in 8085 which determines smallest number in an array of 50 numbers. (CO1, CO2)

3. (a) Explain different types of interrupts of 8086. (CO2, CO3)

(b) Explain relative addressing mode and based + indexed + displacement addressing mode in 8086 with two examples. Also defermine physical memory address in each example. (CO2, CO3)

(c) Explain the meaning of the following 8086 instructions : (CO3, CO5)

(i) XLAT

(ii) STOSB

(iii) AAM

(iv) LAHF

4. (a) Write a program in 8086 to convert a 16-bit binary number into gray number. Also determine addressing mode of each instruction used in this program. (CO3, CO5)

(b) Write a program in 8086 which determine sum of digits of a 16-bit number. (CO3,CO5)

(c) Draw and explain the block diagram of 8254. (CO3, CO5)

5. (a) What is the application of 8251 USART ? Explain its block diagram.

(b) Draw the control word format of 8255 in input-output mode. Write a program in 8085 which reverse contents of port B and store the result at port A and port C. Address of control register is 83H, PA 80H, PB = 81H and PC 82H. (CO6)

(c) Write a program in 8085 which interface ADC 0800 with 8085. (CO6)

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# TCS–404

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

COMPUTER ORGANIZATION

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the unsigned multiplication method (binary multiplication sequencer). Perform the following binary unsigned multiplication operation : (CO1, CO2)

$$(0110)_2 \times (1110)_2$$

(b) Compare the Von Neumann and non-Von Neumann architecture and explain the bottleneck issue in Von Neumann architecture. (CO1, CO2)

(c) Explain the signed integer representation in detail and find whether arithmetic overflow/underflow is present in the following signed operation or not : (CO1, CO2)

(i) $(0011) + (0111)$

(ii) $(1111) + (1101)$

*P. T. O.*

2. (a) Explain Zero, One, Two and Three address assembly language notations for the given arithmetic operation X = (A+B)*C. (CO2, CO3)

   (b) Explain the execution of a complete instruction using single bus structure. (CO2, CO3)

   (c) Differentiate addressability and addressing modes using suitable examples. (CO2, CO3)

3. (a) Compare the memory mapped and non-memory mapped I/O devices. Also discuss the steps of an I/O operation execution. (CO3, CO5)

   (b) What are interrupts ? How are they classified ? Explain all the methods of the interrupt nesting. (CO3, CO5)

   (c) Explain any *two* of the following terms in brief : (CO3, CO5)

      (i) Direct Memory Access and Centralized Bus Arbitration

      (ii) Handshake Control Data Transfer

      (iii) USB Standard I/O Interface

4. (a) Describe memory hierarchy based on parameters like speed, size, and cost. (CO4)

   (b) What is the role of cache memories in computer systems ? Explain the direct mapped cache with suitable diagram. (CO4)

   (c) Explain any *two* of the following terms in brief : (CO4)

      (i) Virtual Memory Paging

      (ii) Effective Access Time

      (iii) Hit Ratio

5 (a) Explain the arithmetic, instruction and processor level pipelining. Also describe the role of cache memory in pipelining. (CO2, CO6)

(b) Describe the Flynn's taxonomy of computer architecture and its various classifications. (CO2, CO6)

(c) Explain any *two* from the following terms in brief : (CO2, CO6)

(i) Shared Memory Processors

(ii) Superscalar and VLIW operations

(iii) Parallel Processor Architecture

Roll No. ..............................

# TCS–408

## B. TECH. (CSE) (FOURTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023

JAVA PROGRAMMING LANGUAGE

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) For Programming questions provide typical output.

1. (a) Explain features of Java that makes it platform independent, portable and robust. Also, create a class with in this package "AmountInwords" to convert amount into words.

(Consider amount not to be more than 100000)

Example :

Typical Input : 3423

Typical Output : Three Four Two Three (CO1, CO2)

(b) Explain the need of StringBuffer class and also write a program in JAVA to convert given Full Name e.g. "Ajay Singh" into First Name i.e. "Ajay" and Last Name i.e."Singh". Additionally, sort the given list of names in ascending order of First Name. (CO1, CO2)

(c) Differentiate between Character class and String class. Explain how tol compare two objects of String class. Also write a program that converts lowercase characters to uppercase characters of a string. (CO1, CO2)

2. (a) "Constructors cannot be static." Justify your answer with suitable programming example. Also, explain that more than one constructor can be defined inside the class with suitable programming example. (CO2)

(b) Explain parameterized constructor and write a Java program which takes two $3 \times 3$ matrices as input and find sum of the two matrices by using constructor for initializing matrix objects. (CO2)

(c) Define the calling of base class constructor inside the derived class. Also, write a Java program to create a Teacher class and derived Professor/Associate_Professor/Assistant_Professor class from Teacher class with appropriate constructor for all the classes. Additionally, define a method to display information of Teacher. (CO2)

**HINT :** Make necessary assumptions as required.

3. (a) Differentiate the terms interface and class. Design an interface called Shape with methods draw() and getArea(). Further, design two classes called Circle and Rectangle that implements Shape to compute area of respective shapes. Also, use appropriate getter and setter methods. (CO3)

(b) Explain how, does Java implements the model of inierprocess synchronization using Threads with the help of a program. Also, write a program that demonstrate the priority setting in threads. (CO3)

(c) Explain the advantages of exception handling and also, explain how exception subclasses are created. Create your own exception subclass "MyEx class" which takes an string as input and throws a message "string too large" when the size of string is more than ten characters.

(CO3)

4. (a) Illustrate the concept of character and byte stream classes in Java. Also, write a program which reads the content of a given file and writes it on the console. (CO4, CO5)

(b) Explain the different types of AWT components. How are these components added to the container ? Also, design a user interface to collect data from the student for admission application using swing components. (CO4, CO5)

(c) Differentiate between AWT and Swings with the help of a Java code. Which is preferred among the two and why ? (CO4, CO5)

5. (a) Explain the uses of ArrayList, HashSet and Linked List with suitable programming example. (CO5, CO6)

(b) Define object class. how this is different from generics in Java ? Also, explain your answer with appropriate programming example that why we should use generic instead of object class. (CO5, CO6)

(c) What are the characteristics of JDBC ? What are the various types of JDBC ? Write a program to demonstrate how JDBC connection is established. (CO5, CO6)

1000

Roll No. ..............................

# TCS–409

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

DESIGN AND ANALYSIS OF ALGORITHMS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Write code in C, C++, Java, or python only.

1. (a) Use the substitution method to prove a tight asymptotic upper bound on the solution to the recurrence : (CO1)

(A) $T(n) = T(n/4) + n^2$

$T(1) = 1$

(B) $T(n) = \begin{cases} 8T\left(\frac{n}{2}\right) + \theta(1), & \text{if } n^2 < M \\ M & \text{, if } n^2 \leq M \end{cases}$

Where M is independent of n.

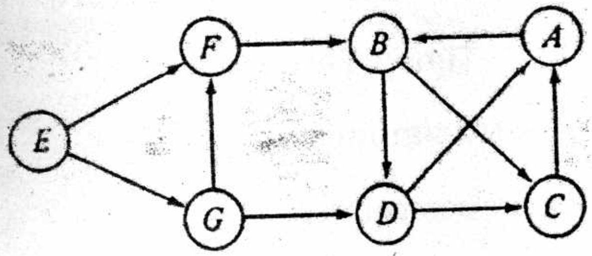
*P. T. O.*

(b) A file contains the following characters and their corresponding frequencies as shown below :

a : 27, e : 35, i : 42, o : 10, u : 35, t : 12

We use Huffman coding for data comparisons, generate the encoding for a, e, i, o, u, t using Huffman encoding and find the average length of a character after compression. (CO3)

(c) Differentiate between DFS and BFS. Apply DFS on the following graph and write minimum hop from source vertex (E) to every other vertex. : (CO2)




2. (a) Prove the following : (CO5)

(i) fib[2n] = fib[n]* (fib[n] + 2fib[n – 1])

(ii) fib[2n – 1] = fib[n – 1]* fib[n – 1] + fib[n] * fib[n]

where fib[x] is $x^{th}$ Fibonacci term fib[0] = 0, fib[1] = 1.

[**Hint** : start with matrix $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \text{fib}[0] & \text{fib}[1] \\ \text{fib}[1] & \text{fib}[2] \end{pmatrix}$ and raise its power.]

(b) Complete the following function using bottom up dynamic programming for 0/1 knapsack : (CO5)

```
int knapsack (int w[], int v[], int n, int k)
{
   /* Complete the function*/
}
```

w is weight array, v is value array n is number of items, k is knapsack capacity.

Fill the dynamic programming table for the following values :

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Weight | 3 | 2 | 1 | 5 | 4 |
| Value | 9 | 6 | 5 | 14 | 4 |

Knapsack capacity = 6

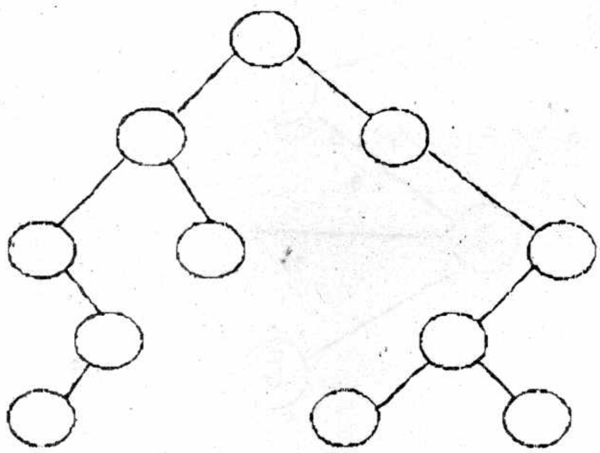
(c) Write the pseudo code of selection sort. (CO4)

What will be the output after 3 iterations of bubble sort on the following numbers ?

5, 7, 6, 4, 3, 9, 0, 2, 1

3. (a) Differentiate between P, NP, NP-Complete, NP-Hard. Write the recursive function for Longest common subsequence. (CO5)

(b) What is Naïve string-matching algorithm ? How is it different from Rabin Karp ? Write code for Rabin Krap. (CO1)



(c) Fill the above tree for BST using the following keys : (CO2)

10, 12, 14, 16, 19, 21, 25, 29, 31, 33, 36

4. (a) Differentiate between Greedy and dynamic programming. Write code for activity selection and matrix chain multiplication. (CO2)

(b) Explain Bellman-ford algorithm with code. Differentiate between Bellman-Ford and Dijkstra. (CO5)

(c) Following keys are inserted in an initially empty max heap in the following order : (CO4)

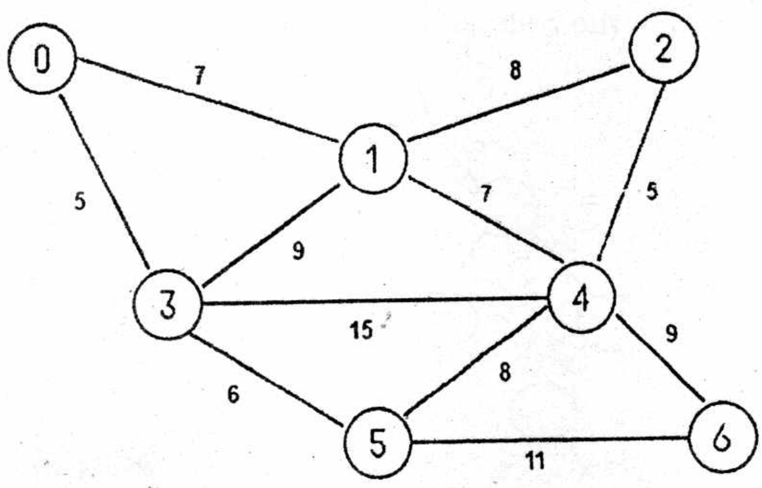
18, 20, 3, 17, 87, 13, 44, 56, 80.

What will be the final max heap ?

5. (a) Differentiate between quick sort and merge sort. Write algorithm of merge sort. (CO1)

(b) What is hashing ? Define collision and different collision handling techniques. (CO4)

(c) Apply Kruskal's algorithm on the following graph, show each step clearly : (CO3)

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# TCS–421

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

FUNDAMENTAL OF STATISTICS AND AI

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Why do we find interquartile ranges ? How are they useful in data exploration phase ? (CO1)

(b) Justify the use of finding Correlation and Regression of the same dataset. (CO1)

(c) "Machine Learning Algorithms plays a powerful role in modern day decision-making." Explain. (CO1)

2. (a) Differentiate between the following : (CO2)

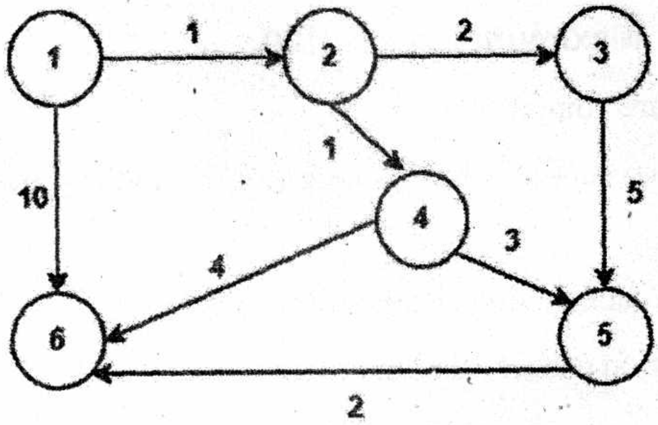
(i) Balanced Data and Unbalanced Data

(ii) Structured Data and Unstructured Data

*P. T. O.*

(b) Discuss A* search algorithm and apply the same for the given graph to search the goal node. Consider Start Node=1 and Goal Node=6 : (CO2)

| | |
|---|---|
| 1 | 5 |
| 2 | 3 |
| 3 | 4 |
| 4 | 2 |
| 5 | 6 |
| 6 | 0 |




(c) Discuss the real-world applications of Data Science. (CO2)

3. (a) What is the difference between covariance and classification ? (CO3)

(b) Discuss the type of hill climbing algorithms and how they are useful. (CO3)

(c) You are given a data set consisting of variables with more than 30 percent missing values. How will you deal with them ? (CO3)

4. (a) "Statistics is the most critical unit of Data Science." Justify the statement in your own words. (CO4)

(b) Consider the student data sets with the following fields :

Id, name, fees, batch, scholarship, dept (Assume the suitable data).

Develop R program to perform following operations :

(i) Get the details of the students belonging to a particular year.

(ii) Get all the students studying in CSE department.

(iii) Get the details of students receiving scholarship. (CO4)

(c) Develop the Python program to display Line chart, Scatter plot and Bar graph. (CO4)

5. (a) Discuss few of the data cleaning activities possible on a real time dataset. (CO5)

(b) Apply rank correlation coefficient formula to find the value of the correlation coefficient from the following data : (CO6)

| X | Y |
|---|---|
| 68 | 62 |
| 64 | 58 |
| 75 | 68 |
| 50 | 45 |
| 64 | 81 |
| 80 | 60 |
| 75 | 68 |
| 40 | 48 |
| 55 | 50 |
| 64 | 70 |

(c) There are two variables Age and Glucose Level, their variable proportions at some discrete timings are given below. Apply Pearson's correlation coefficient formula to find the value of the correlation coefficient : (CO6)

| Age (X) | GL (Y) |
|---|---|
| 43 | 99 |
| 21 | 65 |
| 25 | 79 |
| 42 | 75 |
| 57 | 87 |
| 59 | 81 |



450

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# END SEMESTER EXAMINATION JUNE - 2023

*Name of Program:* **B.Tech** *Semester:* **4th**

*Name of Course:* **Introduction to Machine Learning and Deep Learning**

*Course Code:* **TCS-423**

*Time:* **3 Hours** *Maximum Marks:* **100**

**Note:**
*(i) All questions are compulsory.*
*(ii) Answer any two questions among a, b, & c in each main question.*
*(iii) Total marks in each main question are twenty.*
*(iv) Each question carries 10 marks.*

**Q1. (a) How would you define the feed forward neural network? Elaborate the Layers of feed forward neural networks.** **[CO1, CO4, PO1]**

**(b) How would you present the Gradient Descent algorithm? Mention the Gradient Descent Rules.** **[CO1, CO3, PO3]**

**(c) Can you differentiate the Machine Learning and Deep Learning algorithms in terms of their working and uses?** **[CO6, CO2, PO4]**

**Q2. (a) Elaborate on the support vector machine with an example.** **[CO1, PO5]**

**(b) How would you explain the Naive Bayes Classifier algorithm by using an example.** **[CO3, PO2]**

**(c) Can you illustrate the steps of the ID3 Algorithm. List the capabilities and limitations of the ID3 Algorithm?** **[CO4, PO2]**

Q3. (a) How would you define the AUC-ROC Curve in machine learning? Give an application of AUC-ROC Curve. [CO1, CO2, PO1]

(b) Explain various performance evaluation metrics used for machine learning algorithms. [CO5, CO4, PO2]

(c) Elaborate on bias-variance decomposition. Explain why it is important to find a tradeoff between bias and variance? [CO1, CO3, PO2]

Q4. (a) How can you demonstrate the K-means Clustering Algorithm with an example. [CO2, PO1]

(b) What do you mean by Clustering? Explain with an example. Can you list the categories of clustering algorithms? [CO1, PO2]

(c) Can you elaborate the steps involved in Agglomerative Clustering Algorithm. Demonstrate the steps with the help of an example. [CO1, CO2, PO1]

Q5. (a) How would you define Deep Learning? List some significant applications of deep learning. [CO5, CO6, PO5]

(b) Can you draw and explain the basic architecture of Convolutional Neural Networks. [CO1, CO6, PO1]

(c) Contrast the advantages and disadvantages of deep learning algorithms. [CO6, PO1]

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# TCS–426

## B. TECH. (CSE) (FOURTH SEMESTER) END SEMESTER EXAMINATION, June, 2023

INFORMATION SECURITY RISK MANAGEMENT

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss different types of user authentication techniques. Explain various methods for public key distribution. (CO1)

   (b) Explain symmetric and asymmetric key cryptography, cryptanalysis in brief. (CO1)

   (c) Elaborate the various cryptographic approx. with its mathematical implementation (minimum *three* techniques). (CO2)

2. (a) Suppose that two parties A and B wish to set up a common secret key (D-H key) between themselves using the Diffie-Hellman key exchange technique. They agree on 7 as the modulus and 3 as the primitive root. party A chooses 3 and party B chooses 7 as their respective secrets. Find the key. (CO2)

(b) How to perform risk control strategies in risk management ? (CO3)

(c) Discuss the main building block of project management for information security. (in case of Educational Organisation). (CO3)

3\. (a) Why is risk monitoring required ? Discuss risk monitoring phases with an suitable example. (CO4)

(b) Discuss the steps to for Disaster Recovery Planning (DRP). (CO4)

(c) How can risk monitoring factor affect on an organisation ? Describe with a real time example. (CO4)

4\. (a) Discuss the threat and penetration testing. Explain CK adversary threat model. (CO5)

(b) Explain the different approaches to Information Security Implementation with real-time case study. (CO5)

(c) What is a Cyber Security Audit ? Discuss its benefits for operation security parameters. (CO5)

5\. (a) Explain Common Obligations for Information Security Requirements. (CO6)

(b) Discuss about system security policy model to handle vulnerabilities and security assessment. (CO6)

(c) Define the following (any *four*) : (CO6)

(i) Worms

(ii) Firewal

(iii) Vulnerability Assessment

(iv) Hashing

(v) CIA Theorem

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# TCS–433

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

BLOCKCHAIN AND ITS APPLICATIONS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you mean by Security mechanisms ? Describe all possible types of Security mechanisms with examples. (CO1)

   (b) What do you mean by divisibility ? How is it relevant for deciding primality ? How are prime no's important in Cryptography ? Explain it in brief. (CO1)

   (c) Write stepwise Euclid GCD theorem. What is the relevance of the EUCLID CCD theorem in RSA cryptosystems ? Mention its relevance. (CO2)

2. (a) What do you mean by Additive and multiplicative Cipher ? Mention each of them using examples. (CO2)

   (b) What do you mean by Discrete Logarithm Problem ? Demonstrate the working of the Diffie Hellmann Key Exchange Algorithm stepwise with an example. (CO2)

   (c) What do you by Message Integrity ? How could it be achieved using Digital signature ? Elaborate it with appropriate possible reasons. (CO3)

3. (a) What do you mean by Peer to Peer network ? How is it different from distributed Peer to Peer networks ? Explain it with its diagram. How is it relevant and useful for ensuring support for Blockchain Ecosystem ? (CO3)

   (b) What do you mean by Byzantine Generals Problem ? How is it relevant for the development of the Proof of Byzantine Fault Tolerance Consensus Algorithm for Blockchain ? (CO3)

   (c) Write a bash script to compute the sum of *n* elements taken from command line arguments. (CO4)

4. (a) Discuss the layered architecture of the Blockchain Ecosystem in detail. (CO4)

   (b) Discuss the technical perspective of blockchain with the diagram in detail. (CO5)

   (c) Write the needful steps and commands to make client-server communication using Netcat. (CO5)

5. (a) How can Blockchain facilitate the healthcare ecosystem ? Elaborate with an example and its diagram. (CO5)

   (b) How Security Systems can be enhanced and made decentralized using Blockchain and the same can be a game changer for securing IoT systems ? Explain all major reasons. (CO6)

   (c) What do you mean by Cryptocurrencies ? How can Blockchain enhance the banking system and it could stabilize itself as the Financial infrastructure for the future ? (CO6)

Roll No. ..............................

# TCS–434

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

CLOUD COMPUTING ARCHITECTURE AND DEPLOYMENT MODELS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Syte optimized costs and reduced response times for its visual discovery and automated product tagging solution by migrating its technology stack to AWS and adopting Amazon OpenSearch Service. Evaluate Cloud Computing regularity issues in above mentioned organization.

(CO1)

(b) TUI used AWS to build a branded voting application 90 percent faster, for use by 12,000 audience members on the popular Belgian game show De Mol. Evaluate cloud computing limitations in above mentioned organization.

(CO2)

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(c) Talent acquisition software vendor iCIMS improved performance by migrating nearly 6,000 databases to Amazon RDS for SQL Server. The iCIMS Talent Cloud helps organizations build winning workforces. Evaluate cloud computing security benefits in above mentioned organization. (CO1)

2. (a) Using Amazon DocumentDB for data collection, FINRA has reduced costs month over month by over 50 percent and decreased development times for collecting regulatory filings. Analyze cloud deployment model in above mentioned organization. (CO3)

(b) Using AWS services for its automation solution, cloud security company Botprise saves on operational costs and reduces the time it takes to remediate security issues by 86 percent on average. Analyze PaaS in above mentioned organization. (CO3)

(c) Illumina reduced its carbon emissions by 89 percent by using AWS instead of an on-premises data center and unlocked an additional 60 percent monthly cost savings by using Amazon S3 Intelligent-Tiering. Analyze IaaS in above mentioned organization. (CO3)

3. (a) MediSys Health Network has transformed its IT operations and innovation capabilities by migrating its alternate production environment to AWS. Evaluate cloudSim architecture in above mentioned organization. (CO4)

(b) Tufts Medicine implemented its EHR system on AWS, reducing technical debt, improving security, increasing resiliency, reducing costs, and improving caregiver and patient experience. Evaluate challenges of VMware virtualization in above mentioned organization. (CO5)

(c) Panasonic Avionics Corporation migrated its databases to the cloud to collect, analyze, and store data more efficiently so that it can provide near-real-time data to airlines. Evaluate the steps to virtualize a physical machine in above mentioned organization. (CO4)

4. (a) The Centers for Medicare & Medicaid Services cut log data storage costs by 67 percent using Amazon OpenSearch Service, freeing up taxpayer money to improve efficiency elsewhere around the agency. Evaluate AWS services in above mentioned organization. (CO5)

(b) Dataminr used AWS Inferentia to increase data throughput per dollar by five times. The company can now process four to five times as much data with the same AI models, enhancing its event alerts for global customers. Evaluate Microsoft Azure services in above mentioned organization. (CO4)

(c) Global educational technology company instructure improved its compute throughput by upto 30 percent while decreasing costs by upto 20 percent after migrating to Amazon EC2 instances based on AWS Graviton processors. Evaluate IBM cloud services in above mentioned organization. (CO6)

5. (a) With a framework built on AWS IoT Core, Helen of Troy uses data analysis to bring value to customers with a connected experience, improve its products, and facilitate innovation. Analyze Workload Distribution Architecture in above mentioned organization. (CO3)

(b) CloudCall is facilitating its transformation to a software-as-a-service business model by upskilling its engineers with AWS Training and Certification. Analyze Service Load Balancing Architecture in above mentioned organization. (CO5)

(c) Gaming company Whatwapp scales its business, manages its data and code, and delivers new features to players by using game server Nakama running on Amazon EKS clusters. Analyze Cloud Computing Reference Architecture in above mentioned organization. (CO6)



50

Roll No. ..............................

# TCS–435

## B. TECH. (CSE) (FOURTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023

INTRODUCTION TO AI

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

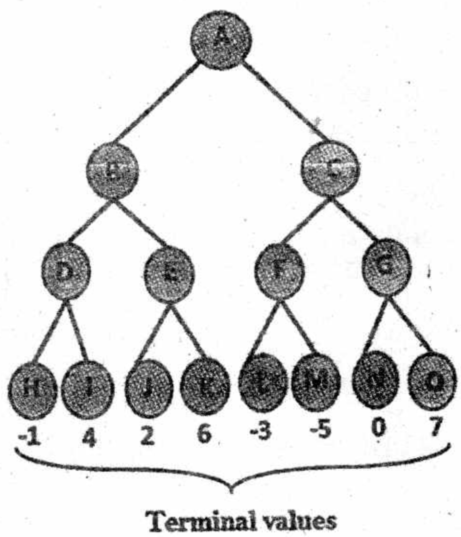
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the concept of hidden layers in a feedforward network. Why are they called "hidden" ? Explain the significance of having multiple hidden layers. (CO3)

(b) By utilizing the min-max algorithm, solve the following search tree : (CO1)

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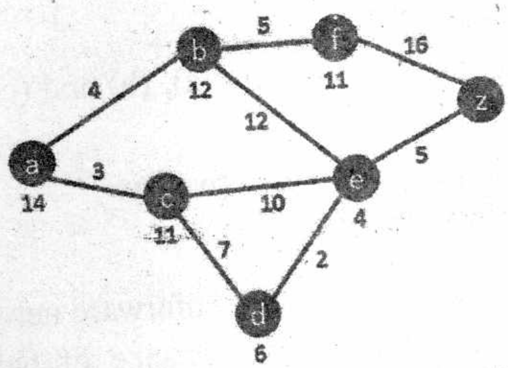
(c) Discuss the future of AI and its potential impact on various industries and society as a whole. (CO1)

2. (a) Demonstrate the forward propagation mechanism in a feedforward network. Explain how the network's output is calculated given an input using suitable mathematics and a block diagram. (CO4)

(b) Elucidate the structure and environment for an intelligent agent. (CO3)

(c) By utilizing the A* search algorithm, solve the following search tree, where *b* is the starting node and *z* is the target node : (CO3)



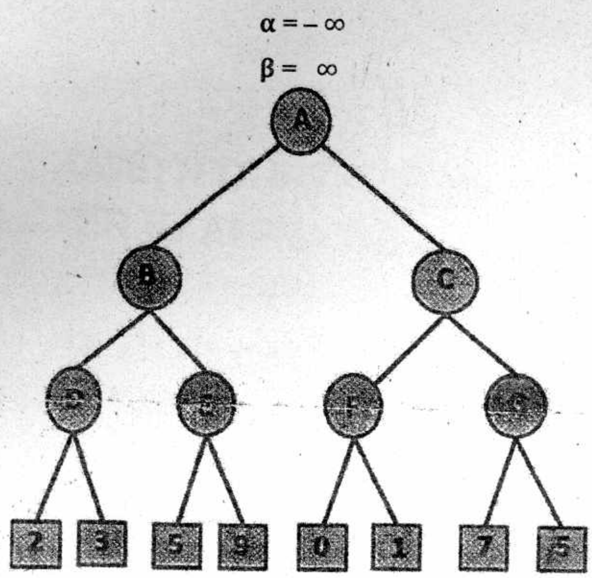

3. (a) Define the knowledge-based agent in an artificial intelligence system. Define the architecture, layers, and methodologies used to create a knowledge-based agent. (CO2)

(b) Explain and draw a block diagram of the Wumpus world theorem. (CO4)

(c) Elaborate the engineering applications of Artiticial Intelligence with proper examples. (CO5)

4. (a) Contrast forward and backward chaining in the context of Artificial Intelligence, emphasizing major distinctions and similarities. (CO4)

(b) Describe the working principles and advantages of the Alpha-Beta pruning algorithm in the context of game tree search and apply it for solving the following search tree : (CO2)




(c) How well does the McCulloch-Pitts neuron model address the concept of learning and adaptation in neural networks ? (CO3)

5. (a) Train the neural network for AND gate by using perception learning method. The initials weights are 0.1 and 0.2. (CO1)

(b) Describe the Propositional logic in Artificial Intelligence with proper examples. (CO2)

(c) Explain the following terms with examples : (CO5)

(i) Depth first search algorithm

(ii) Breadth first search algorithm

(iii) Perceptron learning rule

(iv) Delta learning rule

Roll No. ..............................

# TCS–442

## B. TECH. (AI & DS) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

INTRODUCTION TO DATA MINING AND MACHINE LEARNING

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) By assuming samples that are vector tuples denoting collected data, illustrate at least five different matchers used in data mining. (CO2)

(b) Define the following metrics and illu tı ıte with graphically depict inter-relations between the metrics : (CO3)

(i) Recognition Accuracy

(ii) ROC

(iii) Precision

(iv) Recall

Roll No. ..............................

# TCS–442

## B. TECH. (AI & DS) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

INTRODUCTION TO DATA MINING AND MACHINE LEARNING

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) By assuming samples that are vector tuples denoting collected data, illustrate at least five different matchers used in data mining. (CO2)

(b) Define the following metrics and illu t ıte with graphically depict inter-relations between the metrics : (CO3)

(i) Recognition Accuracy

(ii) ROC

(iii) Precision

(iv) Recall

(c) Consider the following dataset which is for objects on conveyor belt that are either Apples or Banana : Given w = 378, c = 3; predict the class-label using K-NN, for K = 3 and K = 5.

Feature space = {Weight = w, Color = c} and Class-label = {Apple = A, Banana = B} also, color feature can take only 1 out 3 values given by, c = 1, 2 or 3. (CO5)

| w | c | class/label |
|---|---|---|
| 303, 370, 298, 277 | 2, 2, 2, 2 | B, A, B, B |
| 377 299, 383, 374 | 3, 2, 1, 3 | A, B, A, A |
| 303. 309. 359, 366 | 3, 2, 1, 1 | B, B, A, A |
| 311, 302, 373, 305, 371 | 2, 2, 3, 2, 2 | B, B, A, B, A |

2. (a) Describe with block diagram the pipeline of functions from sensing to classification of data. What are the typical performance metrics for classifier block ? (CO5)

(b) Define following terms : Precision, Recall and F-score in case of search or retrieval. Compute these metrics for table below : (CO4)

| | False Positive | False Negative |
|---|---|---|
| True Negative | 9760 | 140 |
| True Positive | 40 | 60 |

(c) By using a real-world example, discuss step-by-step procedure for performing K-means clustering, Also, give key differences if hierarchical clustering was applied to the same dataset.

3. (a) Explain Feature Engineering in Data Mining (DM) model and define what are outliers and where can it be detected in the DM pipeline ? (CO5)

   (b) Outline various outlier analysis methods. Discuss how underfitting and overfitting is influenced by front and of data mining. (CO2)

   (c) Develop the framework of Collaborative filtering as Recommender system. (CO6)

4. (a) Develop procedural steps in PCA and SVD techniques for feature extraction and compare their relevance for dimensionality reduction. (CO3)

   (b) Given covariance matrix, $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; depict its eigen components and give pointwise comments on feature representation using PCA. (CO4)

   (c) Given data matrix A, where, $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$, compute SVD showing relevant steps of left eigen, singular values and right eigen vectors. (CO4)

5. (a) Illustrate the essential steps involved in Bagging and Boosting as ensemble Machine learning methods. (CO5)

   (b) What is Big data and how is it characterized. Outline the Hadoop architecture in Big data computing. (CO6)

   (c) State precisely how statistical learning is defined in the context of automation/machine ? With example and figure illustrate each: Supervised, Unsupervised and Reinforcement type of Machine Learning. (CO3)

Roll No. ..............................

# TCS–451

## B. TECH. (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

VIRTUALIZATION AND CLOUD COMPUTING

**Time : Three Hours**

**Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) The Chicago Lighthouse expanded opportunities for workers with disabilities by pivoting to remote work using AWS while maintaining connectivity, security, and worker productivity. Evaluate different perspectives on CC in above mentioned organization. (CO1)

(b) TiVo is migrating hundreds of APIs to the cloud using AWS. By using Amazon API Gateway and AWS Lambda, the company is achieving improved scalability, consistent performance, and reduced costs. Evaluate characteristics of cloud computing as per NIST in above mentioned organization. (CO2)

(c) 3M applies science in collaborative ways to improve daily lives. Using AWS Application Migration Service, 3M migrated thousands of workloads to AWS, accelerating its digital transformation and driving innovation. Evaluate Web 2.0 in above mentioned organization. (CO1)

2. (a) LaunchDarkly used Amazon Kinesis to build a feature-management solution that scaled to evaluate about 20 trillion feature flags and ingest 250 TB daily while providing 99.999 percent availability. Analyze Storage virtualization in above mentioned organization. (CO3)

(b) Dexatek increased the performance of its Internet of Things (IoT) platform, enhanced security, and lowered management time by migrating to AWS IoT Core. Analyze Data virtualization in above mentioned organization. (CO3)

(c) Using AWS services, global energy company Iberdrola built the Advanced Smart Assistant to help customers reduce smart device energy consumption and better manage grid consumption and capacity. Analyze Memory virtualization in above mentioned organization. (CO3)

3. (a) Learn how Salesforce built a scalable, agile telehealth solution to help remove barriers to healthcare using Amazon Chime SDK. Evaluate Memory architecture emulation in above mentioned organization. (CO4)

(b) Discover how PBS uses AWS to scale its diversity, equity, and inclusion initiatives to reach diverse audiences. Evaluate Operating system emulation in above mentioned organization. (CO5)

(c) Okta is empowering its engineers to resolve outstanding questions faster and accelerating its speed of innovation using the AWS Support App in Slack. Evaluate Resource virtualization in above mentioned organization. (CO4)

4. (a) ArenaNet used AWS solutions to off-load management and to optimize search functionality of the wikis that accompany its most popular massively multiplayer online role-playing game, Guild Wars. Evaluate Elements of Parallel Computing in above mentioned organization. (CO5)

(b) Geospatial web services provider Geo.me opened market opportunities, expanded innovation possibilities, and optimized costs for its customers using Amazon Location Service. Evaluate Laws of Caution in above mentioned organization. (CO4)

(c) Thomson Reuters accelerated the migration of its largest data center, which housed 800 applications and 40,000 workloads, using Amazon DMA. The company moved 200 TB of data in 45 days to meet ambitious business targets. Evaluate Models for Inter-Process Communication in above mentioned organization. (CO6)

5. (a) Boehringer Ingelheim worked alongside AWS to implement a company-wide data transformation initiative that has helped the company derive financial, efficiency, and compliance benefits. Analyze Dynamic Scalability Architecture in above mentioned organization. (CO3)

(b) Using AWS Organizations, global media company Warner Bros. Discovery improved the merger and acquisition process by creating secure, centralized account deployment. Analyze Service Load Balancing Architecture in above mentioned organization. (CO5)

(c) Nationwide Children's Hospital, CalvertHealth, and Rush University System for Health provide seamless patient care using AWS services. Analyze Cloud Computing Reference Architecture in above mentioned organization. (CO6)



80

Roll No. ...............................

# TCS–467

## B. TECH. (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

LAMP TECHNOLOGIES

**Time : Three Hours**

**Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you mean by LAMP stack ? Explain all the components along with the way these components work together to create a web application. (CO1, CO2)

(b) What is the difference between static and dynamic web pages ? Explain the way you can create static and dynamic pages of a website by using various components of the LAMP stack. (CO1, CO2)

(c) What do you mean by the front end and back end ? Which technologies we have discussed in our course are needed to design and develop them ? Explain their objectives and how these interact with each other to create a website.

2. (a) What do you mean by the get and post method in an HTML form ? Create an HTML form having the following elements with mentioned constraints-username (required), email (required), contact (required, the field having a length of 10, should contain only digits), gender (required, radio box, and one of the gender already checked), city (required, drop-down and at-least one option already selected, submit button). (CO1, CO3)

   (b) Briefly explain pop up boxes in JavaScript. Create two JavaScript variables first Name and last Name and get these inputs from the browser which you need to used further to print a message "Welcome first Name last Name to the World of Web." (CO1, CO3)

   (c) Explain some of the date methods of data object in details. Also write a program to check how many days are left for Christmas. (CO1, CO3)

3. (a) Explain some of the sorting methods of associative arrays in PHP ? Write a program to declare an associative array and sort them in ascending order with the help of keys. (CO1, CO4, CO3)

   (b) Describe some of the feature of PHP. Explain the various parameters that you pass to the function for connecting to the database. (CO1, CO4, CO3)

   (c) Explain, how events are handled in JavaScript. Write an event handler program using function to change the background of a division on a single click of a button. (CO1, CO4, CO3)

4. (a) Explain the associative arrays in PHP along with the detailed description of $_GET and $_POST and $_REQUEST global variables ? Also, use a suitable example, explain the way you can extract the data from $_GET and $_POST from server side using PHP.

(CO4, CO5, CO6)

(b) What are cross-cite scripting attacks ? How can cross-cite scripting attacks be prevented on the client slide ? Write a program in PHP to support your answer. HINT : use $_SERVER ['PHP_SELF']

(CO4, CO5, CO6)

(c) Write down the insert, edit and delete query on (Table-Employee) mentioned below and explain the way you will execute with the help php. Also explain difference between include and required for passing the connection string. (CO4, CO5, CO6)

| Employee's Details | | | | | | |
|---|---|---|---|---|---|---|
| Employee_id | Username | Name | Email | Phone No | Gender | Country |
| | | | | | | |

5. (a) Discuss the various permissions used in file handling, also discuss the significances of using fclose() in file handling ? Write a program to put some numbers in a file main.txt, now open a file in read mode and fetch the contents from the file if numbers are even kept in a separate file even.txt and if they are odd put them in a separate file named odd.php.

(CO4, CO5, CO6)

(b) Describe the difference between sessions and cookies. Which one is better and why ? Illustrate with an example. Write a program to enable disable cookies on a single click of a button. (CO4, CO5, CO6)

(c) Explain the concept of AJAX. Write a program to fetch the content of a JSON file named demo.json with the help of AJAX and visualize the database on the client side.

```
{ "stdID" : "0106",
"stdData" : {
                "stdName" : "Mjrwaise",
                 "stdAge" : "23",
                  "stdGender" : "Male",
                   "stdNo" : "12345",
                    }
"stdEdu" : {
                   "stdDept" : "Computer Science",
                    "stdSemester" : "8",
                     "stdMajor" : "web programming"
                      }
}
```

Roll No. ..............................

# TCS–471

## B. TECH. (CSE) (FOURTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

STATISTICAL DATA ANALYSIS WITH R

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Illustrate the concept of univariate, bivariate sampling and re-sampling with a suitable example. (CO1)

   (b) Find the probability of boys and girls in families with three children, assuming equal probabilities for boys and girls. (CO1)

   (c) Three balls are drawn successively from the box containing 6 red balls, 4 white balls and 5 blue balls. Find the probability that they are drawn in the order red, white and blue if each ball is (i) replaced and (ii) not replaced. (CO1)

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(C03, C05)

2. (a) Illustrate the necessary steps of Data Exploration and Preparation. (CO3, CO6)

   (b) The number of failures occurring in a machine of a certain type in a year has a Poisson distribution with mean an in a factory there are ten of these machines. What is (i) the expected total number of failures in the factory in a year and (ii) the probability that there are fewer than two failures in the factory in a year ? (CO3, CO6)

   (c) Create user defined function in R for calculating the sum and average of any number. (CO3, CO6)

3. (a)

```
i = 0
while (i < 6) {
i = i + 1
if (i == 3) {
break
}
print (i)
}
```

   What will be the output of this R program ? Modify this code using for loop, however the semantics of the program will remain the same. (CO3, CO5)

   (b) Create the matrix in R of elements {Jan., Feb., March. Apr., May, June, July, Aug., Sept.,} of 3×3. Show the syntax in R for adding additional rows and column elements one by one {Oct., Nov., Dec.,} for this existing matrix. (CO3, CO5)

(c) emp_id = c (1:5),

emp_name=("Rick", "Dan", "Michelle", "Ryan", "Gray"),

salary=c(623.3, 515.2, 611.0,729.0, 843.25).

Create the data frame of the data given above, find the summary of the data and how to extract specific column from a data frame and to remove rows and columns in a Data Frame. (CO3, CO5)

4. (a) You have a dataset married exam.csv having multiple entries with three columns viz. Roll no, Name, CGPA the data is to be read along with the header, in the second scenario there is delimiter space, in same file. How to import data and handle missing values in R workspace ? (CO2)

(b) Which data object in R is used to store and process categorical data ? How do you find the levels in R for the given data Jazz, Rock, Classic Classic, Pop, Jazz, Rock and Jazz. (CO2)

(c) Compare standard and command packages. How do you save objects in R workspace ? (CO2)

5. (a) Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Apply one-way ANOVA using a 0.05 significance level to the following data : (CO4)

| Food I | Food II | Food III |
|---|---|---|
| 8 | 4 | 11 |
| 12 | 5 | 8 |
| 19 | 4 | 7 |
| 8 | 6 | 13 |
| 6 | 9 | 7 |
| 11 | 7 | 9 |

(b) Consider a predefined dataset in R you have to establish the relationship using multiple regression among "mpg" as a response variable with "disp", "hp" and "wt" as predictor variables. Write the procedure in R to find its relationship and how will you predict having given values disp = 221, hp = 102 and wt = 2.91. (Predicted actual value is not required, your R code is sufficient). (CO4)

(c) Write the R syntax for designing for the given plots and graphs by using suitable examples : (CO4)

(i) Box plot

(ii) Scatter plots

(iii) Bar charts

(iv) Line charts.

# ANOVA TABLE

**The F Distribution**

α = .05

| $df_D$ \ $df_N$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **1** | 161.4 | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 240.5 | 241.9 |
| **2** | 18.51 | 19.00 | 19.16 | 19.25 | 19.30 | 19.33 | 19.35 | 19.37 | 19.38 | 19.40 |
| **3** | 10.13 | 9.55 | 9.28 | 9.12 | 9.01 | 8.94 | 8.89 | 8.85 | 8.81 | 8.79 |
| **4** | 7.71 | 6.94 | 6.59 | 6.39 | 6.26 | 6.16 | 6.09 | 6.04 | 6.00 | 5.96 |
| **5** | 6.61 | 5.79 | 5.41 | 5.19 | 5.05 | 4.95 | 4.88 | 4.82 | 4.77 | 4.74 |
| **6** | 5.99 | 5.14 | 4.76 | 4.53 | 4.39 | 4.28 | 4.21 | 4.15 | 4.10 | 4.06 |
| **7** | 5.59 | 4.74 | 4.35 | 4.12 | 3.97 | 3.87 | 3.79 | 3.73 | 3.68 | 3.64 |
| **8** | 5.32 | 4.46 | 4.07 | 3.84 | 3.69 | 3.58 | 3.50 | 3.44 | 3.39 | 3.35 |
| **9** | 5.12 | 4.26 | 3.86 | 3.63 | 3.48 | 3.37 | 3.29 | 3.23 | 3.18 | 3.14 |
| **10** | 4.96 | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 | 2.98 |
| **11** | 4.84 | 3.98 | 3.59 | 3.36 | 3.20 | 3.09 | 3.01 | 2.95 | 2.90 | 2.85 |
| **12** | 4.75 | 3.89 | 3.49 | 3.26 | 3.11 | 3.00 | 2.91 | 2.85 | 2.80 | 2.75 |
| **13** | 4.67 | 3.81 | 3.41 | 3.18 | 3.03 | 2.92 | 2.83 | 2.77 | 2.71 | 2.67 |
| **14** | 4.60 | 3.74 | 3.34 | 3.11 | 2.96 | 2.85 | 2.76 | 2.70 | 2.65 | 2.60 |
| **15** | 4.54 | 3.68 | 3.29 | 3.06 | 2.90 | 2.79 | 2.71 | 2.64 | 2.59 | 2.54 |
| **16** | 4.49 | 3.63 | 3.24 | 3.01 | 2.85 | 2.74 | 2.66 | 2.59 | 2.54 | 2.49 |
| **17** | 4.45 | 3.59 | 3.20 | 2.96 | 2.81 | 2.70 | 2.61 | 2.55 | 2.49 | 2.45 |
| **18** | 4.41 | 3.55 | 3.16 | 2.93 | 2.77 | 2.66 | 2.58 | 2.51 | 2.46 | 2.41 |
| **19** | 4.38 | 3.52 | 3.13 | 2.90 | 2.74 | 2.63 | 2.54 | 2.48 | 2.42 | 2.38 |
| **20** | 4.35 | 3.49 | 3.10 | 2.87 | 2.71 | 2.60 | 2.51 | 2.45 | 2.39 | 2.35 |
| **21** | 4.32 | 3.47 | 3.07 | 2.84 | 2.68 | 2.57 | 2.49 | 2.42 | 2.37 | 2.32 |
| **22** | 4.30 | 3.44 | 3.05 | 2.82 | 2.66 | 2.55 | 2.46 | 2.40 | 2.34 | 2.30 |
| **23** | 4.28 | 3.42 | 3.03 | 2.80 | 2.64 | 2.53 | 2.44 | 2.37 | 2.32 | 2.27 |
| **24** | 4.26 | 3.40 | 3.01 | 2.78 | 2.62 | 2.51 | 2.42 | 2.36 | 2.30 | 2.25 |
| **25** | 4.24 | 3.39 | 2.99 | 2.76 | 2.60 | 2.49 | 2.40 | 2.34 | 2.28 | 2.24 |
| **26** | 4.23 | 3.37 | 2.98 | 2.74 | 2.59 | 2.47 | 2.39 | 2.32 | 2.27 | 2.22 |
| **27** | 4.21 | 3.35 | 2.96 | 2.73 | 2.57 | 2.46 | 2.37 | 2.31 | 2.25 | 2.20 |
| **28** | 4.20 | 3.34 | 2.95 | 2.71 | 2.56 | 2.45 | 2.36 | 2.29 | 2.24 | 2.19 |
| **29** | 4.18 | 3.33 | 2.93 | 2.70 | 2.55 | 2.43 | 2.35 | 2.28 | 2.22 | 2.18 |
| **30** | 4.17 | 3.32 | 2.92 | 2.69 | 2.53 | 2.42 | 2.33 | 2.27 | 2.21 | 2.16 |
| **40** | 4.08 | 3.23 | 2.84 | 2.61 | 2.45 | 2.34 | 2.25 | 2.18 | 2.12 | 2.08 |
| **60** | 4.00 | 3.15 | 2.76 | 2.53 | 2.37 | 2.25 | 2.17 | 2.10 | 2.04 | 1.99 |
| **120** | 3.92 | 3.07 | 2.68 | 2.45 | 2.29 | 2.17 | 2.09 | 2.02 | 1.96 | 1.91 |
| **∞** | 3.84 | 3.00 | 2.60 | 2.37 | 2.21 | 2.10 | 2.01 | 1.94 | 1.88 | 1.83 |



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# TCS–472

## B. TECH. (AI & ML) (FOURTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023

INTRODUCTION TO AI AND NEURAL NETWORK

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elucidate the concept of a neural network and its basic components. Discuss the role of activation functions in a neural network and provide examples of commonly used activation functions. (CO3)

(b) What is the role of neural networks in AI ? Explain the structure and functioning of a basic neural network. (CO1)

(c) Discuss the future of AI and its potential impact on various industries and society as a whole. (CO1)

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2. (a) Explain the process of forward propagation in a feedforward network. How is the output of the network computed given an input ? Describe with proper equations and block diagram. (CO4)

   (b) Elucidate the structure and environment for an intelligent agent. (CO3)

   (c) Discuss the role of weights and biases in a feed forward network. How are they adjusted during the training process ? Explain with proper algorithm. (CO3)

3. (a) Elucidate the knowledge-based agent in an AI system. Also describe the architecture, levels, and approaches to design a knowledge, based agent. (CO2)

   (b) Describe the Wumpus world theorem with proper explanation and block diagram. (CO4)

   (c) Elaborateon the role of First Order Logic (FOL) in knowledge engineering and expert systems. Discuss how FOL is utilized to represent and reason about complex knowledge domains, providing examples to support your discussion. (CO5)

4. (a) Compare and contrast forward chaining and backward chaining in the context of artificialintelligence, highlighting their key differences and similarities. (CO4)

   (b) Describe the working principles and advantages of the Alpha-Beta pruning algorithm in the context of game tree search. (CO2)

   (c) What arethe advantages and disadvantages of the McCulloch-Pitts neuron model compared to other computational models of neurons ? (CO3)

5. (a) Train theneural network for OR gate by using perception learning method. The initials weights are 1 and 2. (CO1)

   (b) Describe the A* search algorithm with proper explanation. (CO2)

   (c) Explain the following terms with examples : (CO5)

      (i) Uninformed search technique

      (ii) Informed search technique

      (iii) Artificial neural network

      (iv) Delta learning rule

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# TCS–491

## B. TECH. (CSE) (FOURTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023

INTRODUCTION TO CRYPTOGRAPHY

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss about Active attack and Passive attack. What is masquerade in cryptography ? (CO1)

   (b) What are the essential elements of the RC4 algorithm ? What Networking applications use the RC4 stream cipher ? (CO1)

   (c) Discuss all the versions have X.509 certificates. (CO2)

2. (a) With regard to the first step of processing in each round of AES on the encryption side. How does one look up the 16 x 16 S-box table for byte-by-byte substitutions ? In other words, assuming I want a substitute byte for the byte $b_7\, b_6\, b_5\, b_4\, b_3\, b_2\, b_1\, b_0$ where each $b_i$ is a single bit, how do a user can access these bits to find the replacement byte in the S-box table ? (CO2)

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(b) Discuss about Transparent Key Control Scheme in terms of data security from host to host over the TCP. (CO3)

(c) Elaborate Euler's Totient Theorem. What is the outcome of $3^{2012}$ mod 17 using the same technique ? (CO3)

3. (a) Find the greatest common divisor of 44 and 17, using Euclidean algorithm. Find whole numbers $x$ and $y$ so that $44x + 17y = 1$ with $x > 10$ using GCD. (CO4)

(b) Discuss the role of Intruders and Intrusion Detection System (IDs) in cyber-security. (CO4)

(c) Find integers $x$ and $y$ to satisfy $42823x + 6409y = 17$ using Multiplicative Inverse Algorithm. (CO4)

4. (a) What do you mean by a Phishing attack ? Discuss all types of Phishing attacks. What are the measures for protecting the information against Phishing attacks ? (CO5)

(b) Differentiate RSA and Diffie-Hellman approaches. Write down the main difference in both techniques. (CO5)

(c) What do you mean by DDos Attack ? Discuss all the steps required to perform this attack using the block diagram. (CO5)

5. (a) Find the multiplicative inverse of 8 mod 11, using the Euclidean algorithm. (CO6)

(b) What do you mean by Cyber Security Threat ? What are the different types of cyber security threats possible ? List down them and describe it with its bad effect. (CO6)

(c) Define the following (any *four*) : (CO6)

(i) Worms

(ii) Firewall

(iii) Law Enforcement Challenges

(iv) MD5 Hash Algorithm

(v) Digital Millennium Copyright Act